

Solar-Powered Kennel Light

T.K. HAREENDRAN

Here is the circuit of a solar-powered kennel light. The solar cell panel included gives a voltage that fits the voltage demands of the rechargeable battery cells for back up. The rechargeable cells are charged by solar panel during daytime and these provide power supply for the kennel light at night.

The central part of this compact design is a popular 8-pin DC-DC converter chip MC34063A, which is a monolithic control circuit that contains the primary functions required for DC-to-DC converters. The chip (available in a PDIP-8 package) has an internal temperature-compensated reference, a comparator, controlled duty-cycle oscillator with an active current-limit circuit, driver and high-current output switch.

Here, MC34063A IC is configured to drive a bunch of series-connected ultra-bright white light emitting diodes (LEDs) at a constant current rate. When sufficient sunlight is available, the solar panel charges the rechargeable cells directly, and the solar panel voltage is sampled by the front-end circuitry to disable the LED driver circuit.

When the voltage of the solar panel drops due to low sunlight, the LED driver circuit is enabled automatically to keep the LEDs turned on. Fig. 1 shows the author's prototype.

Circuit and working

The circuit diagram of the sun-powered kennel light is shown in Fig. 2. It

is built around DC-to-DC converter MC34063A (IC1), six 5mm bright white LEDs (LED1 through LED6), three Schottky SR206 diodes (D1 through D3) and a few other components.

The backup battery pack (BATT.1) of this circuit is a series of two 1.2V/500mAh Ni-MH cells. When sunlight is available, this battery pack is charged by the 6V/100mA solar panel connected across CON1 through the 47-ohm (R1) charging current-limiting resistor.

MC34063A IC is configured as a DC-DC boost converter to drive the LEDs (LED1-LED6) using 3V DC supply provided by the battery pack. The 47-ohm resistor R2 limits the operating current of the string of LEDs to a safe value.

When the solar panel is working,

all LEDs (LED1 through LED6) are turned off. However, when the solar panel is switched off, all LED lights are turned on. SPDT switch S1 is used to switch on or switch off the LEDs.

Construction and testing

An actual-size, single-side PCB for the solar-powered kennel light is shown in Fig. 3 and its component layout in Fig. 4. Enclose the PCB in a small box in such a way that you can install it at a suitable place.

The 10k potmeter VR1 is used to sample the DC voltage output from the solar panel. If the solar panel gives enough output during daytime, a fraction of this voltage is fed to pin 5 of IC1 via SR206 diode (D2) to disable IC1. An initial (one-time) setup is required to fine-tune this automatic switching option, which is described below:

Connect a fully-charged 2.4V Ni-MH battery pack to the circuit. Remove the solar panel and switch on switch S1. Ensure that LEDs (LED1 through LED6) are lit. Set the potmeter (VR1) at its mechanical centre position. Apply 6V DC supply to the solar panel input (CON1) from your lab's power supply. Slowly turn the potmeter and stop it when the LED lights go off. Remove your lab power supply, reconnect the solar panel. Verify the test point table voltages to ensure

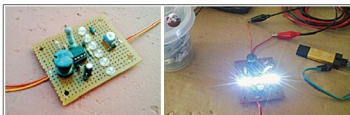


Fig. 1: Author's prototype

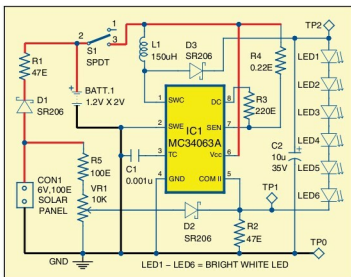


Fig. 2: Circuit diagram of the solar-powered kennel light